Example 1.4

Suppose an electrical equipment manufacturing industry receives screws in lots, as raw materials. The production engineer decides to reject a lot when the number of defective screws is one or more in a randomly selected sample of size 2.

$$\text{Define } X_i = \begin{cases} 1, & \text{if } i^{\text{th}} \text{ screw is defective} \\ 0, & \text{if } i^{\text{th}} \text{ screw is not defective} \end{cases}, \qquad i = 1, 2$$

Then, X_1 and X_2 are *iid* random variables and they have the *Bernoulli* (P) distribution.

Let $H_0 : P = \frac{1}{3}$ and $H_1 : P = \frac{2}{3}$

The sample space is $\mathbf{S}$ = {(0,0),(0,1),(1,0),(1,1)}

If $T(X_1, X_2)$ represents the number of defective screws, in each random sample, then the statistic $T(X_1, X_2) = X_1 + X_2$ is a random variable distributed according to the *Binomial* (2, P) distribution. The possible values of $T(X_1, X_2)$ are 0, 1 and 2. The values of $T(X_1, X_2)$ which lead to rejection of H_0 constitute the set {1,2}.

But, the critical region is defined by the elements of $\mathbf{S}$ corresponding to $T(X_1,X_2) = 1$ or 2. Thus, the critical region is {(0,1), (1,0), (1,1)} whose dimension is 2.

Note 8: When the sampling distribution is continuous, the set of values of $t(\underset{\sim}{X})$ corresponding to the rejection rule will be an interval or union of intervals depending on the alternative hypothesis. It is empahazized that **these intervals identify the elements of critical region, but they do not constitute the critical region**.

When the sampling distribution of the test statistic Z is a normal distribution, the critical values for testing H_0 against the possible alternative hypothesis at two different levels of significance, say 5% and 1% are displayed in Table 1.6.

Table 1.6 Critical values of the Z statistic

Alternative hypothesis	Level of Significance (α)	
	0.05 or 5%	0.01 or 1%
One- sided (right)	$z_\alpha = z_{0.05} = 1.645$	$z_\alpha = z_{0.01} = 2.33$
One- sided (left)	$-z_\alpha = -z_{0.05} = -1.645$	$-z_\alpha = -z_{0.01} = -2.33$
Two-sided	$z_{\alpha/2} = z_{0.025} = 1.96$	$z_{\alpha/2} = z_{0.005} = 2.58$

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